(A) Native-pixel observations

RGB frame / 1920 × 1200



4 × 3 overlapping windows

12 overlapping tiles / 640 × 640



Area-based labels; fruit-free tile sampling

(B) Shared RGB detector

Original



Shared weights

(same encoder–decoder)

Horizontal flip



RT-DETR-L backbone

Multi-scale features

P3

P5

Hybrid encoder

Decoder + detection heads

(C) Instance aggregation

Original view



Internal-edge filtering

Full-frame coordinates

Per-view NMS

Candidates (original)



Mirrored view



Internal-edge filtering

Full-frame coordinates

Per-view NMS

Candidates (mirrored)



Mirror fusion (MF)

Aggregated candidates



(D) Evidence calibration (EC)

Candidate from detector (in full-frame coordinates)



39 candidate features



Score + context

Box + appearance

Depth + curvature

Detector weights frozen during calibration

Correctness estimator BCE

Localization quality Squared loss

Score fusion

$s = p^{(1-\alpha)}u^{\alpha}$

$\alpha = 0.5$

Full-frame detections



Validation labels



Fit after detector training

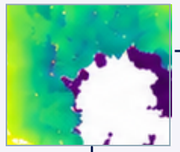
(E) Aligned geometry control

Separate comparison model

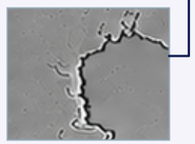
RGB (colour)



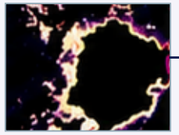
Depth (0.5–6 m)



Convexity map B

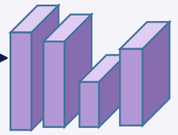


Bumpiness map N



Synchronized transforms

GeoSphere-DETR control



Same tiles and output processing

Inference flow

Calibration supervision

Photo insets illustrate the workflow; feature stacks are schematic.